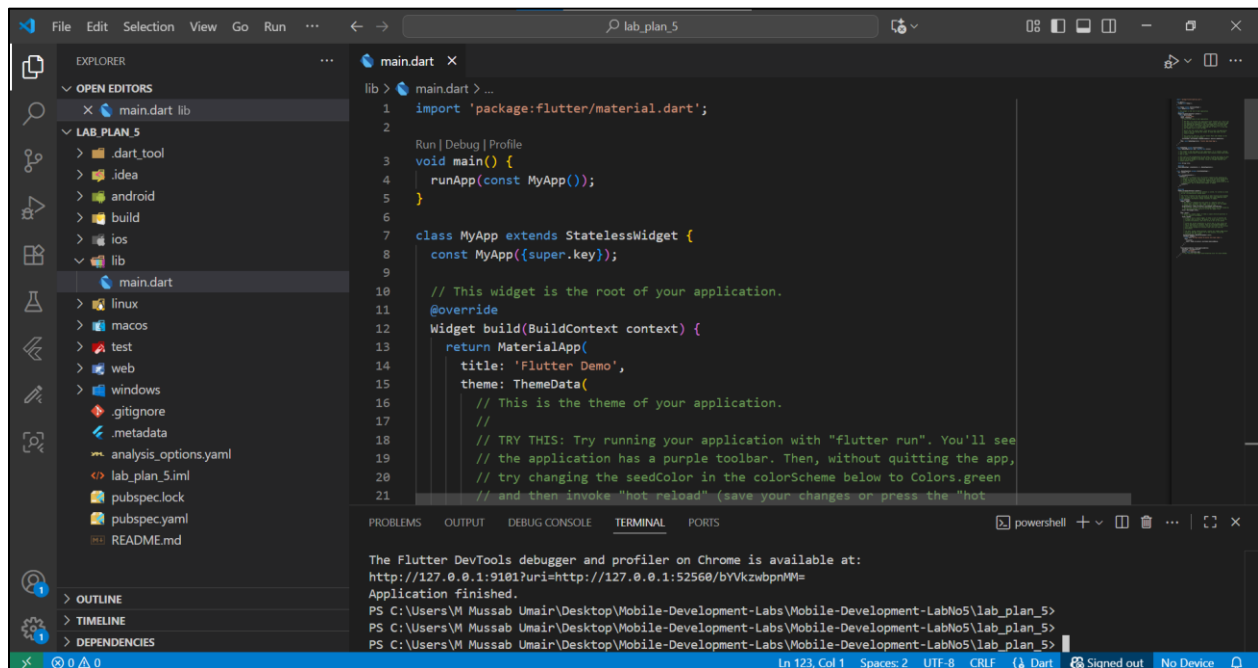


Lab Plan – 5: Flutter ListView, ListTile, SnackBar and Cards

Task 1: Create a Flutter Project

- Open VS Code or Android Studio.
- Use shortcut:
Ctrl + Shift + P → Flutter: New Project (for VS Code).
- Name your project **lab_plan_5** and set up the folder location.



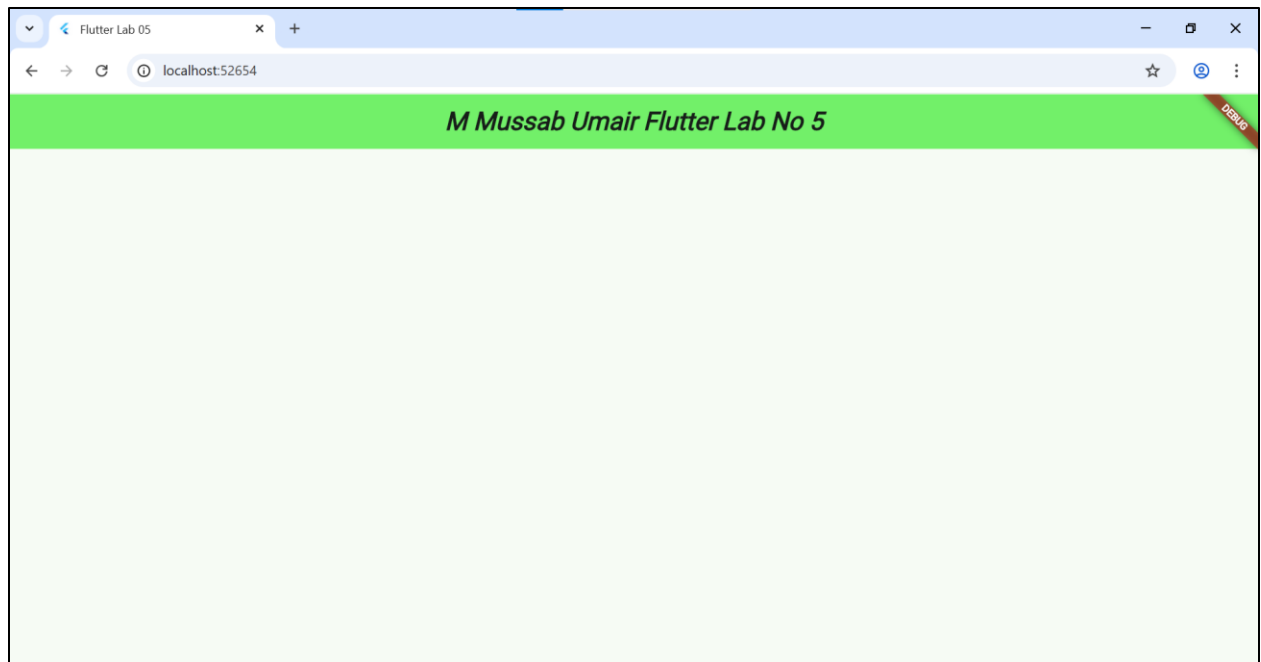
Task 2: Add an AppBar with Centered Text

- Create a Stateless Widget using shortcut **stle**.
- Inside Scaffold, add an **AppBar**.
- Use `centerTitle: true` to align the text in the center.
- Add **font size, font weight, and style** to the AppBar title.

Code:

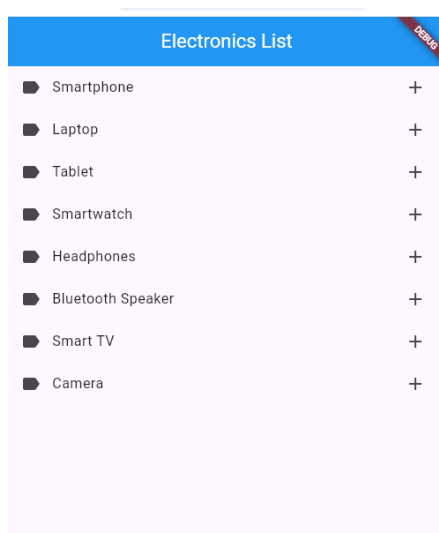
```
• import 'package:flutter/material.dart';
•
• void main() {
•   runApp(const MyApp());
• }
•
• class MyApp extends StatelessWidget {
•   const MyApp({super.key});
•
•   @override
•   Widget build(BuildContext context) {
•     return MaterialApp(
•       title: 'Flutter Lab 05',
•       theme: ThemeData(
•         colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
•         useMaterial3: true,
•       ),
•       home: Scaffold(
•         appBar: AppBar(
•           centerTitle: true,
•           title: const Text(
•             'M Mussab Umair Flutter Lab No 5',
•             style: TextStyle(
•               fontWeight: FontWeight.bold,
•               fontSize: 26,
•               fontStyle: FontStyle.italic,
•             ),
•           ),
•           backgroundColor: const Color.fromARGB(255, 114, 240, 105),
•         ),
•       ),
•     );
•   }
• }
```

Screenshot:



Task 3: Display Items with ListView and ListTile

- Use a **ListView** widget in the body of Scaffold.
- Each item should be displayed using a **ListTile** widget.
- The title of ListTile should be the name of the item.



Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Lab 05',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.greenAccent),
        useMaterial3: true,
      ),
      home: Scaffold(
        appBar: AppBar(
          centerTitle: true,
          title: const Text(
            'M Mussab Umair Electronics',
            style: TextStyle(
              fontWeight: FontWeight.bold,
              fontSize: 26,
              color: Colors.white,
              letterSpacing: 2,
            ),
          ),
        ),
        backgroundColor: const Color.fromARGB(255, 240, 166, 105),
        body: ListView(
          children: const [
            ListTile(
              tileColor: Color.fromARGB(255, 102, 105, 110),
              leading: Icon(Icons.phone_android),
              title: Text('Smartphone'),
              subtitle: Text('Samsung, Oppo, Vivo, Realme'),
              trailing: Icon(Icons.favorite),
              iconColor: Colors.orange,
            ),
          ],
        ),
      ),
    );
  }
}
```

```
),  
);  
}  
}
```

Screenshot:



Task 4: Create a List of Items (Choose icons of your choice)

- Smartphone
- Laptop
- Tablet
- Smartwatch
- Headphones
- Desktop Computer
- Wireless Earbuds
- Gaming Console
- Smart TV
- Bluetooth Speaker
- Power Bank

- Portable Hard Drive
- Drone
- Digital Camera
- Keyboard
- Mouse

Note: if you want to create a snackbar in the same widget then use a Builder class

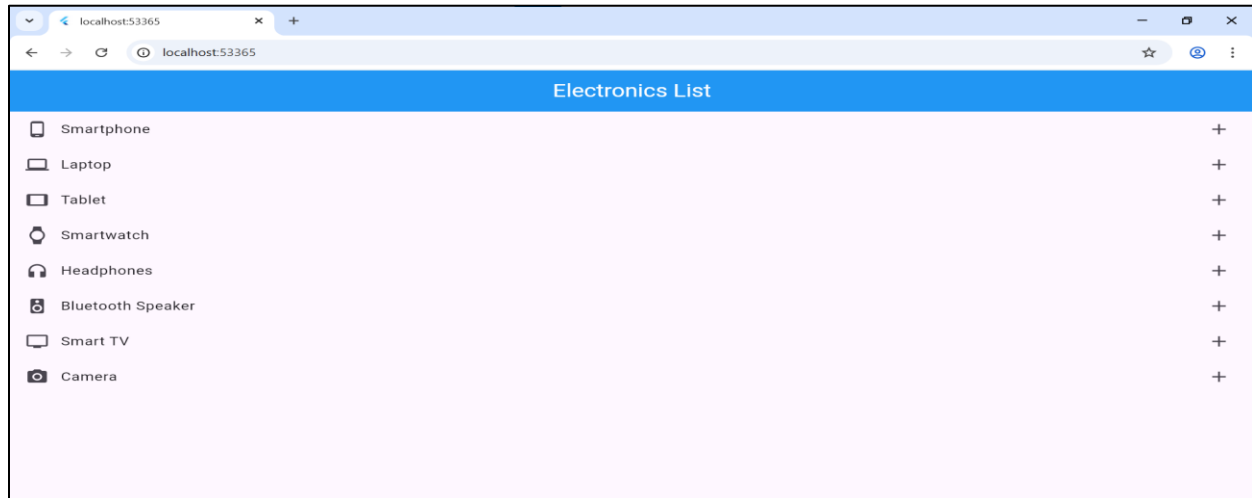
```
Scaffold(  
  body: Builder(  
    builder: (context) => ListView(  
      children: [  
        ListTile(  
          onTap: () {  
            ScaffoldMessenger.of(context).showSnackBar(...);  
          },  
        ),  
      ],  
    ),  
  ),  
);
```

Code:

```
import 'package:flutter/material.dart';  
  
void main() {  
  runApp(  
    MaterialApp(  
      debugShowCheckedModeBanner: false,  
      home: Scaffold(  
        appBar: AppBar(  
          title: const Text(  
            'Electronics List',  
            style: TextStyle(fontSize: 24, color: Colors.white),  
          ),  
          centerTitle: true,
```

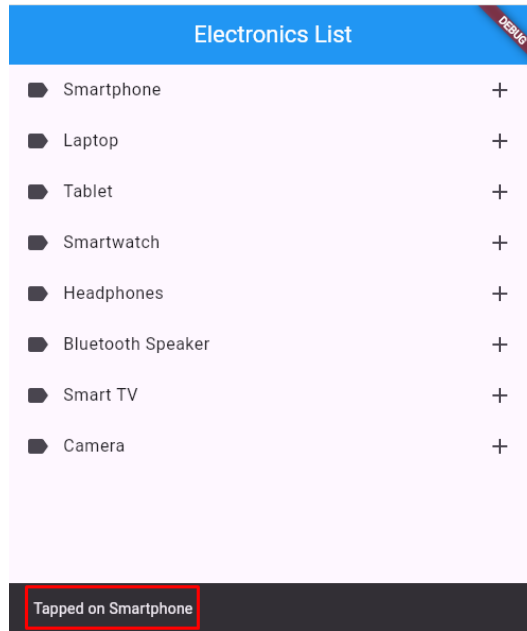
```
        backgroundColor: Colors.blue,
      ),
      body: ListView(
        children: const [
          ListTile(
            leading: Icon(Icons.phone_android),
            title: Text('Smartphone'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.laptop),
            title: Text('Laptop'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.tablet),
            title: Text('Tablet'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.watch),
            title: Text('Smartwatch'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.headphones),
            title: Text('Headphones'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.speaker),
            title: Text('Bluetooth Speaker'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.tv),
            title: Text('Smart TV'),
            trailing: Icon(Icons.add),
          ),
          ListTile(
            leading: Icon(Icons.camera_alt),
            title: Text('Camera'),
            trailing: Icon(Icons.add),
          ),
        ],
      ),
    ),
  ),
),
```

```
    ),  
    ),  
    ),  
    );  
}
```



Task 5: Add Snackbar

- Use the **onTap** property of ListTile.
- When a user taps an item, display a **Snackbar** at the bottom.
- The Snackbar should show a message like:
"Tapped on Smartphone"
- The item name should change dynamically depending on which item is tapped.



Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(
    MaterialApp(
      debugShowCheckedModeBanner: false,
      home: Scaffold(
        appBar: AppBar(
          title: const Text(
            'Electronics List',
            style: TextStyle(fontSize: 24, color: Colors.white),
          ),
          centerTitle: true,
          backgroundColor: Colors.blue,
        ),
        body: Builder(
          builder: (context) => ListView(
            children: [
              ListTile(
                leading: Icon(Icons.phone_android),
                title: Text('Smartphone'),
                trailing: Icon(Icons.add),
                onTap: () {
```

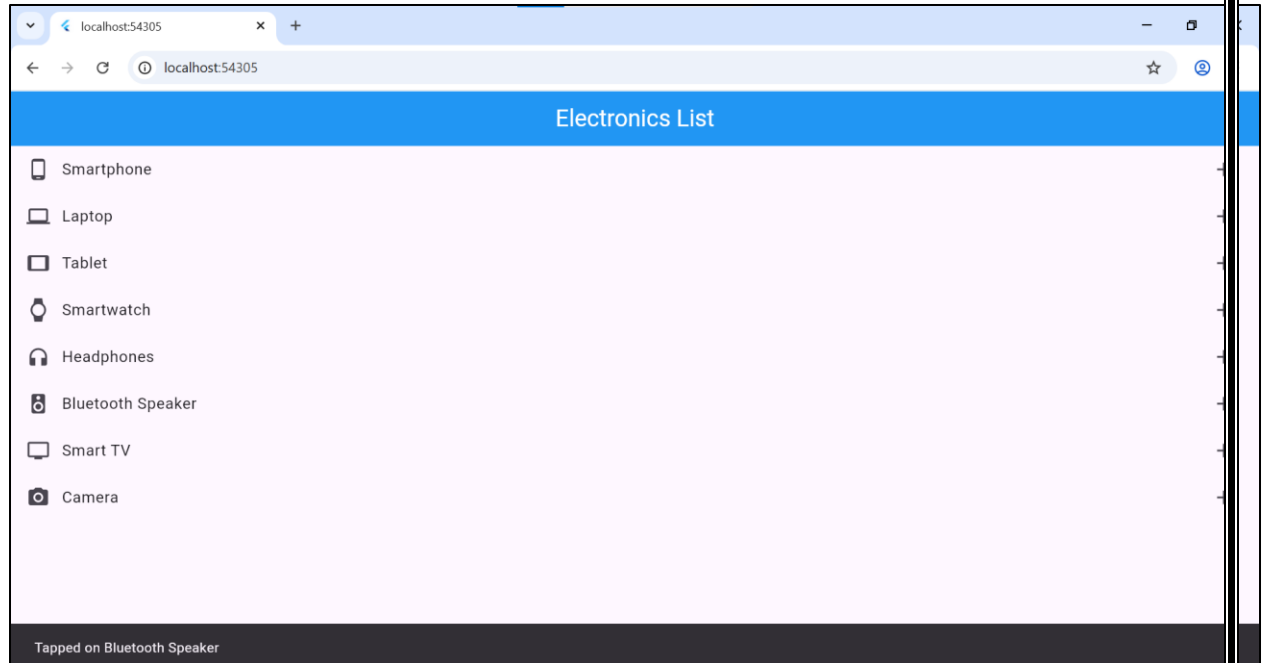
```
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text('Tapped on Smartphone')),
        );
      },
    ),
    ListTile(
      leading: Icon(Icons.laptop),
      title: Text('Laptop'),
      trailing: Icon(Icons.add),
      onTap: () {
        ScaffoldMessenger.of(
          context,
        ).showSnackBar(SnackBar(content: Text('Tapped on Laptop')));
      },
    ),
    ListTile(
      leading: Icon(Icons.tablet),
      title: Text('Tablet'),
      trailing: Icon(Icons.add),
      onTap: () {
        ScaffoldMessenger.of(
          context,
        ).showSnackBar(SnackBar(content: Text('Tapped on Tablet')));
      },
    ),
    ListTile(
      leading: Icon(Icons.watch),
      title: Text('Smartwatch'),
      trailing: Icon(Icons.add),
      onTap: () {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text('Tapped on Smartwatch')),
        );
      },
    ),
    ListTile(
      leading: Icon(Icons.headphones),
      title: Text('Headphones'),
      trailing: Icon(Icons.add),
      onTap: () {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text('Tapped on Headphones')),
        );
      },
    ),
  ),
```

```

ListTile(
  leading: Icon(Icons.speaker),
  title: Text('Bluetooth Speaker'),
  trailing: Icon(Icons.add),
  onTap: () {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text('Tapped on Bluetooth Speaker')),
    );
  },
),
ListTile(
  leading: Icon(Icons.tv),
  title: Text('Smart TV'),
  trailing: Icon(Icons.add),
  onTap: () {
    ScaffoldMessenger.of(
      context,
    ).showSnackBar(SnackBar(content: Text('Tapped on Smart TV')));
  },
),
ListTile(
  leading: Icon(Icons.camera_alt),
  title: Text('Camera'),
  trailing: Icon(Icons.add),
  onTap: () {
    ScaffoldMessenger.of(
      context,
    ).showSnackBar(SnackBar(content: Text('Tapped on Camera')));
  },
),
],
),
),
),
),
);
}

```

Screenshot:



Task 6: Create Flutter Cards

Create a Flutter app that displays a list of electronic products using Card widgets. Each card should show:

- An icon or image
- The product name
- A short description
- A price tag

 **Smartphone**
Powerful Android device with 8GB RAM
 **Price: \$799**

 **Laptop**
High performance for work and gaming
 **Price: \$1299**

 **Smartwatch**
Stay connected on the go
 **Price: \$299**


```

        color: Colors.blue,
      ),
    ),
  ],
),
 SizedBox(height: 8),
 RichText(
    text: TextSpan(
      style: TextStyle(fontSize: 16, color: Colors.black),
      children: [
        TextSpan(text: 'Powerful '),
        TextSpan(
          text: 'Android ',
          style: TextStyle(color: Colors.blue),
        ),
        TextSpan(text: 'device '),
        TextSpan(
          text: 'with ',
          style: TextStyle(color: Colors.purple),
        ),
        TextSpan(
          text: '8GB ',
          style: TextStyle(color: Colors.orange),
        ),
        TextSpan(text: 'RAM'),
      ],
    ),
  ),
  SizedBox(height: 8),
  Row(
    children: [
      Icon(
        Icons.monetization_on,
        color: Colors.orange,
        size: 20,
      ),
      SizedBox(width: 4),
      Text(
        'Price: \$799',
        style: TextStyle(
          fontSize: 16,
          fontWeight: FontWeight.w600,
        ),
      ),
    ],
  ),
],

```

```
),  
    ],  
  ),  
),  
Card(  
  elevation: 4,  
  margin: const EdgeInsets.only(bottom: 16),  
  child: Padding(  
    padding: const EdgeInsets.all(16.0),  
    child: Column(  
      crossAxisAlignment: CrossAxisAlignment.start,  
      children: [  
        Row(  
          children: [  
            Icon(Icons.laptop, size: 30, color: Colors.grey[700]),  
            SizedBox(width: 16),  
            Text(  
              'Laptop',  
              style: TextStyle(  
                fontSize: 20,  
                fontWeight: FontWeight.bold,  
              ),  
            ),  
          ],  
        ),  
        SizedBox(height: 8),  
        RichText(  
          text: TextSpan(  
            style: TextStyle(fontSize: 16, color: Colors.black),  
            children: [  
              TextSpan(text: 'High '),  
              TextSpan(  
                text: 'performance ',  
                style: TextStyle(color: Colors.blue),  
              ),  
              TextSpan(text: 'for '),  
              TextSpan(  
                text: 'work ',  
                style: TextStyle(color: Colors.purple),  
              ),  
              TextSpan(text: 'and '),  
              TextSpan(  
                text: 'gaming',  
                style: TextStyle(color: Colors.purple),
```

```
        ),  
      ],  
    ),  
  ),  
  SizedBox(height: 8),  
  Row(  
    children: [  
      Icon(  
        Icons.monetization_on,  
        color: Colors.orange,  
        size: 20,  
      ),  
      SizedBox(width: 4),  
      Text(  
        'Price: \$1299',  
        style: TextStyle(  
          fontSize: 16,  
          fontWeight: FontWeight.w600,  
          color: Colors.blue,  
        ),  
      ),  
    ],  
  ),  
],  
),  
],  
),  
),  
Card(  
  elevation: 4,  
  margin: const EdgeInsets.only(bottom: 16),  
  child: Padding(  
    padding: const EdgeInsets.all(16.0),  
    child: Column(  
      crossAxisAlignment: CrossAxisAlignment.start,  
      children: [  
        Row(  
          children: [  
            Icon(Icons.watch, size: 30, color: Colors.black),  
            SizedBox(width: 16),  
            Text(  
              'Smartwatch',  
              style: TextStyle(  
                fontSize: 20,  
                fontWeight: FontWeight.bold,  
              ),  
            ),
```



```
),
],
),
 SizedBox(height: 8),
 RichText(
   text: TextSpan(
     style: TextStyle(fontSize: 16, color: Colors.black),
     children: [
       TextSpan(text: 'Stay '),
       TextSpan(text: 'connected '),
       TextSpan(
         text: 'on ',
         style: TextStyle(color: Colors.purple),
       ),
       TextSpan(text: 'the '),
       TextSpan(
         text: 'go',
         style: TextStyle(color: Colors.purple),
       ),
     ],
   ),
 ),
 ),
 SizedBox(height: 8),
 Row(
   children: [
     Icon(
       Icons.monetization_on,
       color: Colors.orange,
       size: 20,
     ),
     SizedBox(width: 4),
     Text(
       'Price: \$299',
       style: TextStyle(
         fontSize: 16,
         fontWeight: FontWeight.w600,
         color: Colors.blue,
       ),
     ),
   ],
 ),
 ],
 ),
 ],
 ),
 ),
 ),
```

```

Card(
  elevation: 4,
  margin: const EdgeInsets.only(bottom: 16),
  child: Padding(
    padding: const EdgeInsets.all(16.0),
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        Row(
          children: [
            Icon(Icons.tablet, size: 30, color: Colors.grey[600]),
            SizedBox(width: 16),
            Text(
              'Tablet',
              style: TextStyle(
                fontSize: 20,
                fontWeight: FontWeight.bold,
              ),
            ),
          ],
        ),
        SizedBox(height: 8),
        Text(
          'Portable entertainment and productivity device',
          style: TextStyle(fontSize: 16, color: Colors.black87),
        ),
        SizedBox(height: 8),
        Row(
          children: [
            Icon(
              Icons.monetization_on,
              color: Colors.orange,
              size: 20,
            ),
            SizedBox(width: 4),
            Text(
              'Price: \$399',
              style: TextStyle(
                fontSize: 16,
                fontWeight: FontWeight.w600,
              ),
            ),
          ],
        ),
      ],
    ),
  ],
),
],

```



Screenshot:

